Is there an enzyme for nylon-6,6?

Querying 33,246 ModelSEED reactions with the encoded diad-forming reaction

8 October 2026

ECFP4 + DRFP reaction embedding · ModelSEED master @ `b539adb6`

The question

Nylon-6,6 is a polyamide: **adipate** (C6 diacid) + **hexamethylenediamine** (C6 diamine), joined by amide bonds.

Is there anything in ModelSEED that does this chemistry — and what enzyme catalyses it?

Both monomers are already ModelSEED compounds

cpd03642 adipate · cpd25003 1,6-diaminohexane · cpd09842 adipyl-CoA

The substrates exist. Only the amide-forming *reaction* is missing.

Approach: encode the target reaction in the same embedding the EC-prediction benchmark selected (ECFP4+DRFP), find its neighbours, then screen those neighbours by *mechanism* before trusting any ranking.

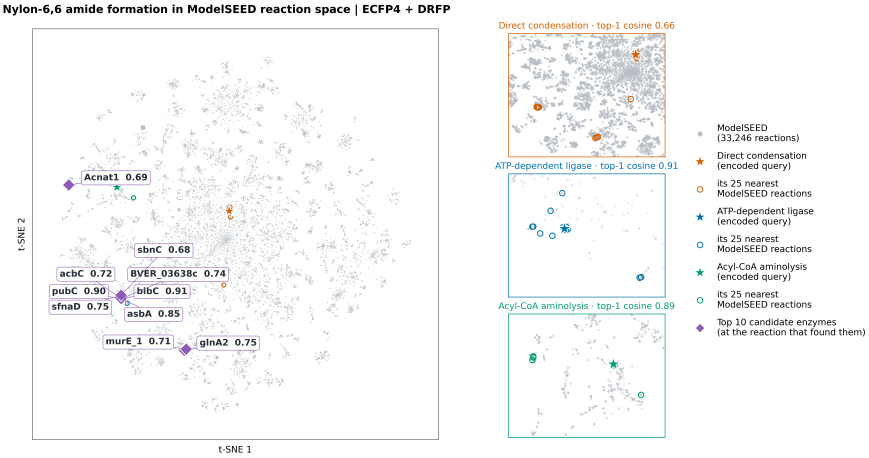
Biology does not condense a free acid with a free amine

So the target is encoded in **six mechanistic framings**, each a separate query, each asserted mass- and charge-balanced.

Framing	Top-1 cosine	Out of distribution?
Direct condensation (acid + amine)	0.502	yes (4.9th pct)
ATP $\rightarrow$ AMP + PPi	0.751	no
ATP $\rightarrow$ ADP + P_i	0.749	no
Acyl-CoA aminolysis	**0.890**	no

The nylon step is unprecedented *only in its uncatalysed form*. Posed the way an enzyme would do it — activate the acid, then attack with the amine — it lands inside populated ModelSEED territory.

Where the target lands in reaction space



Diamonds are the top candidate enzymes, drawn at the reaction that retrieved them. t-SNE does not preserve cosine rank — the ranking is authoritative, the map is for orientation.

Cosine alone is not evidence of chemistry

A whole-reaction difference fingerprint is dominated by the largest compositional change — **the cofactor**, not the amide bond.

Worked example

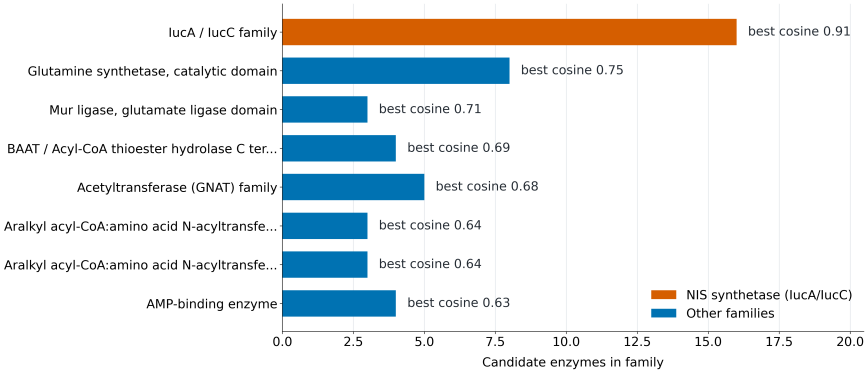
The acyl-CoA query's nearest EC-labelled neighbours are thioester hydrolases (EC 3.1.2, agreement 0.80, no abstain). They make **no amide at all**; they merely release CoA.

Candidates therefore pass three **embedding-independent** screens, applied *before* any top-N cut:

1. a free amine is acylated — proven by **cutting the product amide and reconstituting the parent amine**, which also *names* the acceptor;
2. that nitrogen is a genuine aliphatic amine (not aromatic, amide, nitrile or guanidino);
3. the substrate is a small molecule, not a protein.

33,246 reactions → **726** acylate a free amine → **482** eligible after excluding macromolecular (224, less 5 polymer-elongation cases re-admitted) and unbalanced (22). **24 curated screen controls** plus 327 unit tests guard this.

The hits converge on one family



IucA/IucC — the NIS (NRPS-independent siderophore) synthetases — is recovered 41 times over by independent seeds. One family found forty-one times is convergence, not forty-one independent hits.

Why this family: they make polyamides

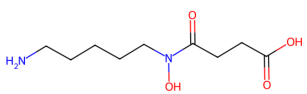
NIS synthetases form ATP-dependent amide bonds between carboxylic acids and **free polyamines** — and several **iteratively oligomerise** bifunctional monomers into dimers and trimers.

Enzyme	Native reaction (UniProt, experimental)	Cosine	Family
BibC	2 × succinyl-N-OH-cadaverine + 2 ATP → bisucaberin	**0.911**	**IucA/IucC**
PubC	2 × succinyl-N-OH-putrescine + 2 ATP → putrebactin	**0.898**	**IucA/IucC**
AsbA	spermidine + citrate + ATP → N^8-citryl-spermidine	0.851	**IucA/IucC**
AsbB	N^8-citryl-spermidine + spermidine + ATP (chain extension)	0.772	**IucA/IucC**
SfnaD	D-ornithine + citrate + ATP	0.753	**IucA/IucC**
PuuA	putrescine + L-glutamate + ATP → γ-Glu-putrescine	0.749	Gln synthetase

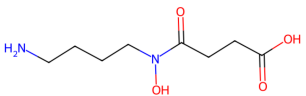
BibC and PubC take *one* bifunctional monomer and join it head-to-tail **twice**. That is polycondensation — the same operation nylon-6,6 synthesis requires.

The native substrate is the nylon diad's structural class

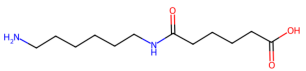
BibC native monomer
succinyl-N-hydroxy-cadaverine (C4+C5)



PubC native monomer
succinyl-N-hydroxy-putrescine (C4+C4)



Nylon-6,6 diad (this work)
adipoyl-hexamethylenediamine (C6+C6)



All three are **diamine–diacid half-amides**: free carboxylate at one end, free primary amine at the other. BibC and PubC natively *dimerise* this exact motif.

This reframes the experiment: not only "make the diad", but **feed the pre-formed diad to BibC/PubC** and look for dimerisation — the chain-extension step of nylon-6,6. Gap: native monomers are N-hydroxylated and two carbons shorter per arm.

Candidates for the bench

Enzyme	UniProt	Family	aa	Why test it
PubC	P0DXD6	**IucA/IucC**	629	cleanest dimerisation of a half-amide
BibC	B6EP94	**IucA/IucC**	819	same, C5 monomer; highest cosine
AsbA	A0A0F7RJ52	**IucA/IucC**	602	two *different* monomers (AA/BB)
AsbB	Q81RQ8	**IucA/IucC**	612	chain extension; has a PDB structure
PuuA	P78061	Gln synthetase	472	*E. coli*; easiest expression/assay

Deliverable

`results/shortlist_annotated.tsv` — a **301-protein** bench shortlist over **50 families** and 205 organisms, drawn from **54,199** candidates (2,048 Swiss-Prot reviewed). Carries **253 AlphaFold models** (median pLDDT **93.8**, none unusable), 105 experimental structures, 41 cofactor-bound, **194 proteins with a measured** K_M/k_{cat}, and 48 with **measured nylon-6,6 titre**.

Similarity is to *reactions*. No listed enzyme is known to accept adipate or HMDA — this analysis **generates** that hypothesis, it does not validate it.

ModelSEED has no reaction → enzyme mapping

So breadth comes from having **seven independent bridges**, not one good one. Every row records which produced it.

Route	What it claims	Candidates
`reaction`	annotated to a screened ModelSEED reaction	816
`precomputed_link`	a global UniProt→ModelSEED table	2,554
`domain`	Pfam/InterPro family with *characteristic* chemistry	9,365
`clade`	UniRef50 neighbourhood of a validated protein	36,889
`enzyme_name`	UniProt *names* it for the target function	4,512
`assayed`	**measured nylon-6,6 titre**	49
`proxied`	literature-curated, carries a geometric score	14

ModelSEED is not the bottleneck. Screening all 36,552 Rhea reactions with our own screen found **417** amide-forming reactions unreachable from ModelSEED — and they yield 3 proteins, because 21 of the top 25 have *no enzyme annotated at all*. The missing links do not exist.

A score must rank retrieval, never gate it

Cosine on a whole-reaction fingerprint is dominated by the cofactor, yet it was deciding *which reactions were ever looked at* — in five places, including one this project added itself.

Gate	What it discarded	Cost
`TOPN=30` per query	41 of the 62 best-scoring reactions	high
`priority` tie-break	broke every tie on cosine	now **zero**
OOD floor on admission	96 of 205 reactions, the *highest*-scoring	all FPGS, all Mur
stage 04 re-litigating stage 03	`rxn14123` at cosine 0.499 vs floor 0.5041	126 proteins
our own `SD_FLOOR=0.40`	214 of 482 *screen-proven* reactions	see below

Removing our own floor: reactions resolved 168 $\rightarrow$ **308**, candidates $+2{,}021$ — and annotation coverage went *up*. Gating on a score discards evidence; ranking on it does not.

Judge the family, not just the protein

A protein admitted by family membership **inherits** its family's score, so the decision governing tens of thousands of candidates is made once per family. It is now made explicitly, on the *measured* chemistry of each family's own validated seeds.

Verdict	n	Criterion
lead	54	diacid + diamine, self-condensing, or bifunctional one side
watch	16	monofunctional, or the donor drags a large scaffold
reject	33	terminal amide, or the members contradict the chemistry

What seed chemistry alone misses

γ-glutamyltranspeptidase scores 0.64 on its seeds and looked like a lead — while 1,550 of its 1,594 members are annotated as a non-amide-forming activity. It is a hydrolase family running a transpeptidation side reaction.

103 families from 298 Pfam/InterPro signatures; the verdict **prunes 8,409** candidates from the bench list. This is the lever that makes the search more *targeted*, not larger.

What we found along the way

The 2026-10-06 ModelSEED structure refresh is wrong for amides

∼17% of compounds — including NAD, NADH, NADP, CoA, every acyl-CoA, nicotinamide — write carboxamides as iminols (`C([O-])=N` and `C(O)=N`), with adenine N1 as `[nH+]`. An iminol is **invisible to an amide SMARTS**. 6,313 compounds repaired; the repair reproduces the previous snapshot exactly.

Honest limitations

- The repair is *not* charge-neutral (5,807 compounds shift charge), so the corpus audits balance separately; unbalanced reactions are excluded, not hidden.
- BibC and PubC carry **no Rhea and no EC in ModelSEED**; both were recovered by pivoting through **MetaNetX**. 308 of 482 screened reactions resolve to an enzyme.
- Screens are structural inference, not atom-mapped proof.

Reproducible by `run_all.py`: 23 stages, ∼40 min, mostly polite API querying; 327 unit tests, 24 screen controls.